

Small Oscillations

1 One Degree of Freedom

Consider a single generalized coordinate q obeying $\ddot{q} = f(q)$. An equilibrium configuration $q = q_0$ is defined by

$$0 = f(q_0). \quad (1)$$

To study the motion for a small displacement away from q_0 , write

$$q = q_0 + \eta, \quad \eta \text{ "small"}, \quad (2)$$

so that $(q_0 + \eta) = \ddot{\eta}$, while Taylor expanding the force about the equilibrium point gives

$$f(q_0 + \eta) = \underbrace{f(q_0)}_{=0} + \eta f'(q_0) + \dots \quad (3)$$

Hence, to leading order in the small deviation,

$$\ddot{\eta} \approx f'(q_0) \eta \quad (\text{the equation of motion is linear in small deviations}). \quad (4)$$

Two behaviors are possible depending on the sign of $f'(q_0)$:

$$\text{Oscillation} \iff f'(q_0) < 0, \quad \omega = \sqrt{-f'(q_0)}, \quad \eta = \eta_0 \cos(\omega t + \varphi_0), \quad (5)$$

$$\text{Instability} \iff f'(q_0) > 0, \quad \lambda = \sqrt{f'(q_0)}, \quad \eta = Ae^{\lambda t} + Be^{-\lambda t}. \quad (6)$$

2 Several Degrees of Freedom

2.1 Equilibrium and the Potential Expansion

For several degrees of freedom, restrict to the case $V = V(q_1, \dots, q_n, t)$, so that

$$L = T - V. \quad (7)$$

An equilibrium configuration $(q_1, \dots, q_n) = (q_{01}, \dots, q_{0n})$ is one at which every generalized force vanishes,

$$Q_j = - \left. \frac{\partial V}{\partial q_j} \right|_{\vec{q}=\vec{q}_0} = 0. \quad (8)$$

Figure 1 illustrates the idea: an equilibrium point is a local minimum of $V(q_j)$, and near that minimum the potential is well approximated by a quadratic form in the displacement η_j .

Writing $q_j = q_{0j} + \eta_j$ and Taylor expanding V to second order in η ,

$$V(q_1, \dots, q_n) = \underbrace{V(q_{01}, \dots, q_{0n})}_{\text{constant} \Rightarrow \text{no worries}} + \sum_{j=1}^n \underbrace{\left. \frac{\partial V}{\partial q_j} \right|_{\vec{q}_0}}_{-Q_j=0} \eta_j + \frac{1}{2} \sum_{i,j=1}^n \underbrace{\left. \frac{\partial^2 V}{\partial q_i \partial q_j} \right|_{\vec{q}_0}}_{\equiv V_{ij}} \eta_i \eta_j + \dots, \quad (9)$$

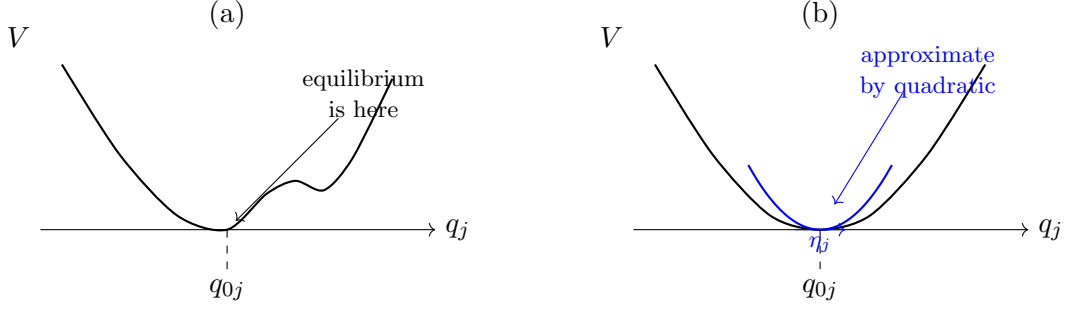


Figure 1: (a) A generic potential $V(q_j)$ with an equilibrium (local minimum) at $q_j = q_{0j}$. (b) Near the equilibrium point, $V(q_j)$ is approximated by a quadratic (parabolic) form in the small displacement η_j .

where V_{ij} is an $n \times n$ matrix with $V_{ij} = V_{ji}$.

Analogously, the kinetic energy is quadratic in the generalized velocities to leading order,

$$T = \frac{1}{2} \sum_{i,j=1}^n \left. \frac{\partial^2 T}{\partial \dot{\eta}_i \partial \dot{\eta}_j} \right|_{\dot{\eta}=0} \dot{\eta}_i \dot{\eta}_j, \quad T_{ij} \equiv \left. \frac{\partial^2 T}{\partial \dot{\eta}_i \partial \dot{\eta}_j} \right|_{\dot{\eta}=0}, \quad T_{ij} = T_{ji}. \quad (10)$$

Explicitly, for a system of N particles,

$$T = \sum_{p=1}^N \frac{m_p}{2} \left(\dot{\vec{r}}_p \right)^2, \quad \dot{\vec{r}}_p = \sum_{j=1}^n \frac{\partial \vec{r}_p}{\partial q_j} \dot{q}_j + \frac{\partial \vec{r}_p}{\partial t}, \quad \text{assuming } \frac{\partial \vec{r}_p}{\partial t} = 0. \quad (11)$$

2.2 Lagrangian and Equations of Motion

Combining Eqs. (9)–(10), the Lagrangian for small oscillations about equilibrium is

$$L = \frac{1}{2} \sum_{i,j=1}^n (T_{ij} \dot{\eta}_i \dot{\eta}_j - V_{ij} \eta_i \eta_j). \quad (12)$$

As an illustration of the derivatives involved, for two degrees of freedom

$$\frac{\partial}{\partial \eta_2} \left(\frac{1}{2} [V_{11} \eta_1 \eta_1 + V_{12} \eta_1 \eta_2 + V_{21} \eta_2 \eta_1 + V_{22} \eta_2 \eta_2] \right) = \frac{1}{2} [(V_{12} + V_{21}) \eta_1 + 2V_{22} \eta_2]$$

$V_{12} \equiv V_{21} \quad V_{21} \eta_1 + V_{22} \eta_2,$

where the $\frac{1}{2}$ prefactor cancels with the factor of 2 produced by the derivative. In general,

$$\frac{\partial L}{\partial \eta_i} = - \sum_{j=1}^n V_{ij} \eta_j, \quad p_i = \frac{\partial L}{\partial \dot{\eta}_i} = \sum_{j=1}^n T_{ij} \dot{\eta}_j. \quad (13)$$

For a single degree of freedom this reproduces the familiar simple harmonic oscillator: with $T = \frac{m}{2} \dot{\eta}^2$ and $V = \frac{k}{2} \eta^2$,

$$L = \frac{m}{2} \dot{\eta}^2 - \frac{k}{2} \eta^2, \quad \frac{\partial L}{\partial \eta} = -k\eta, \quad \frac{\partial L}{\partial \dot{\eta}} = m\dot{\eta}, \quad (14)$$

$$0 = \frac{\partial L}{\partial \eta} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\eta}} \right) = -k\eta - m\ddot{\eta} \implies m\ddot{\eta} = -k\eta. \quad (15)$$

The Lagrange equations following from Eq. (12) are $0 = \partial L / \partial \eta_i - \dot{p}_i$, i.e.,

$$0 = \sum_{j=1}^n V_{ij} \eta_j + \sum_{j=1}^n T_{ij} \ddot{\eta}_j, \quad i = 1, \dots, n. \quad (16)$$

2.3 Normal-Mode Ansatz and the Secular Equation

Propose a common-frequency oscillatory solution,

$$\eta_j = \eta_{j0} \cos(\omega t + \delta_j) \implies \ddot{\eta}_j = -\omega^2 \eta_j, \quad j = 1, \dots, n. \quad (17)$$

Substituting into Eq. (16),

$$0 = \sum_{j=1}^n (V_{ij} \eta_j - \omega^2 T_{ij} \eta_j) = \sum_{j=1}^n (V_{ij} - \omega^2 T_{ij}) \eta_{j0} \cos(\omega t + \delta_j), \quad i = 1, \dots, n. \quad (18)$$

Since $\cos(\omega t + \delta_j) \neq 0$ for most times, this can only hold if

$$0 = \sum_{j=1}^n \underbrace{(V_{ij} - \omega^2 T_{ij})}_{\text{eigenvalue}} \underbrace{\eta_{j0}}_{\text{eigenvector}}, \quad i = 1, \dots, n, \quad (19)$$

with ω the normal-mode frequency and η_{j0} the normal mode. For the η_{j0} to be nontrivial, we need

$$\text{Det} [V_{ij} - \omega^2 T_{ij}] = 0. \quad (20)$$

Aside. More details on the kinetic energy term. Writing out the sum in Eq. (11),

$$T = \sum_{p=1}^N \frac{m_p}{2} (\dot{\vec{r}}_p)^2 = \sum_{p=1}^N \frac{m_p}{2} \left(\sum_{i=1}^n \frac{\partial \vec{r}_p}{\partial q_i} \dot{q}_i + \frac{\partial \vec{r}_p}{\partial t} \right) \cdot \left(\sum_{j=1}^n \frac{\partial \vec{r}_p}{\partial q_j} \dot{q}_j + \frac{\partial \vec{r}_p}{\partial t} \right), \quad (21)$$

with $\vec{r}_p = \vec{r}_p(q_1, \dots, q_n, t) \Rightarrow \dot{\vec{r}}_p = \sum_{j=1}^n \frac{\partial \vec{r}_p}{\partial q_j} \dot{q}_j + \frac{\partial \vec{r}_p}{\partial t}$. Collecting terms by power of the generalized velocities,

$$T = T_2 + T_1 + T_0, \quad (22)$$

$$T_2 = \frac{1}{2} \sum_{i,j=1}^n m_{ij} \dot{q}_i \dot{q}_j, \quad m_{ij} \equiv \sum_{p=1}^N m_p \frac{\partial \vec{r}_p}{\partial q_i} \cdot \frac{\partial \vec{r}_p}{\partial q_j}, \quad (23)$$

$$T_1 = \sum_{j=1}^n \pi_j \dot{q}_j, \quad \pi_j \equiv \sum_{p=1}^N m_p \frac{\partial \vec{r}_p}{\partial q_j} \cdot \frac{\partial \vec{r}_p}{\partial t}, \quad T_0 = \sum_{p=1}^N \frac{m_p}{2} \left(\frac{\partial \vec{r}_p}{\partial t} \right)^2 \quad (24)$$

(m_{ij} is a function of $(q_1, \dots, q_n)$ because the derivatives $\partial \vec{r}_p / \partial q_k$ depend on $(q_1, \dots, q_n)$). Assuming $\partial \vec{r}_p / \partial t = \vec{0}$ and expanding for $q_j = q_{0j} + \eta_j$ with η_j small gives

$$T = T_2 = \frac{1}{2} \sum_{i,j=1}^n \left(m_{ij}|_{\vec{q}_0} + \sum_k \frac{\partial m_{ij}}{\partial q_k} \Big|_{\vec{q}_0} \eta_k + \dots \right) (\dot{q}_{0i} + \dot{\eta}_i)(\dot{q}_{0j} + \dot{\eta}_j). \quad (25)$$

But $(q_{0i} + \eta_i)' = \dot{\eta}_i$, and $\eta_k \dot{\eta}_i \dot{\eta}_j \propto \eta^3$ is of higher order in the small deviations; therefore, up to quadratic order in $\eta, \dot{\eta}$,

$$T = \frac{1}{2} \sum_{i,j=1}^n T_{ij} \dot{\eta}_i \dot{\eta}_j, \quad \text{with} \quad T_{ij} = m_{ij}|_{\vec{q}=\vec{q}_0}. \quad (26)$$

2.4 General Solution and Normal Coordinates

Equation (20) has, in general, n roots ω_k^2 ($k = 1, \dots, n$, the number of degrees of freedom),

$$\text{Det} [V_{ij} - \omega_k^2 T_{ij}] = 0, \quad [\mathbf{V} - \omega_k^2 \mathbf{T}] \vec{a}_k = \vec{0}, \quad k = 1, \dots, n. \quad (27)$$

The general solution is a linear combination of the normal modes,

$$\eta_i = \sum_k C_k a_{ik} \cos(\omega_k t + \delta_k), \quad \vec{\eta} = \sum_k C_k \vec{a}_k \cos(\omega_k t + \delta_k). \quad (28)$$

Defining the matrix of eigenvectors $(\mathbf{A})_{ij} \equiv a_{ij}$, $\mathbf{A} \equiv (\vec{a}_1 \ \vec{a}_2 \ \dots \ \vec{a}_n)$, one has the property

$$\mathbf{1} = \mathbf{A}^T \mathbf{T} \mathbf{A} \implies \mathbf{A}^{-1} = \mathbf{A}^T \mathbf{T}, \quad \mathbf{A}^T \mathbf{V} \mathbf{A} = \begin{pmatrix} \omega_1^2 & 0 & \dots \\ 0 & \ddots & 0 \\ \vdots & 0 & \omega_n^2 \end{pmatrix}. \quad (29)$$

Normal modes example. Figure 2 shows three equal masses connected in a line by identical springs. The three normal modes (Fig. 3) are $\vec{a}_1 = (1, 0, -1)^T$ at frequency ω_1 , $\vec{a}_2 = (-1, 2, -1)^T$ at frequency ω_2 , and $\vec{a}_3 = (1, 1, 1)^T$ at $\omega_3 = 0$. The general motion is

$$\begin{pmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \end{pmatrix} = C_1 \vec{a}_1 \cos(\omega_1 t + \delta_1) + C_2 \vec{a}_2 \cos(\omega_2 t + \delta_2) + C_3 \vec{a}_3 \cos(\omega_3 t + \delta_3). \quad (30)$$

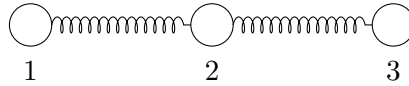


Figure 2: Three equal masses coupled in a line by identical springs.

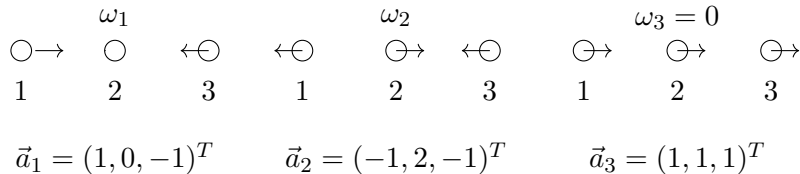


Figure 3: The three normal modes of the system in Fig. 2: $\vec{a}_1$ at ω_1 , with the outer masses displaced oppositely and the middle mass fixed; $\vec{a}_2$ at ω_2 , with the middle mass displaced oppositely to, and at twice the amplitude of, the outer masses; and $\vec{a}_3$ at $\omega_3 = 0$, a uniform translation of the chain.

Introducing normal coordinates $\vec{\zeta}$ through $\vec{\eta} = \mathbf{A}\vec{\zeta}$, i.e. $\eta_i = \sum_k \zeta_k a_{ik}$ with $\zeta_k = C_k \cos(\omega_k t + \delta_k)$, so that $\vec{\zeta} = \mathbf{A}^{-1}\vec{\eta} = \mathbf{A}^T \mathbf{T} \vec{\eta}$, the Lagrangian becomes

$$L = \frac{1}{2} \sum_{i,j} (\dot{\eta}_i T_{ij} \dot{\eta}_j - \eta_i V_{ij} \eta_j) = \frac{1}{2} [\dot{\vec{\eta}}^T \mathbf{T} \dot{\vec{\eta}} - \vec{\eta}^T \mathbf{V} \vec{\eta}] = \frac{1}{2} [\dot{\vec{\zeta}}^T \mathbf{A}^T \mathbf{T} \mathbf{A} \dot{\vec{\zeta}} - \vec{\zeta}^T \mathbf{A}^T \mathbf{V} \mathbf{A} \vec{\zeta}], \quad (31)$$

$$L = \frac{1}{2} \left[\dot{\vec{\zeta}}^T \dot{\vec{\zeta}} - \vec{\zeta}^T \begin{pmatrix} \omega_1^2 & & 0 \\ & \ddots & \\ 0 & & \omega_n^2 \end{pmatrix} \vec{\zeta} \right] \Rightarrow \boxed{L = \frac{1}{2} \sum_{k=1}^n (\dot{\zeta}_k^2 - \omega_k^2 \zeta_k^2)}. \quad (32)$$

In the normal coordinates the system decouples into n independent harmonic oscillators.

3 Example: Oscillations of a Double Pendulum

3.1 Setup and Lagrangian

Consider two point masses m connected in series by rigid, massless rods of length ℓ , as in Fig. 4, with angles φ_1, φ_2 measured from the vertical.

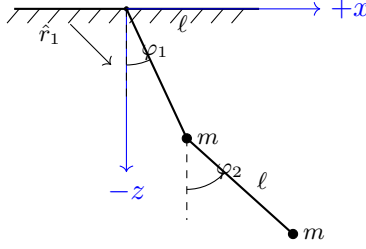


Figure 4: Double pendulum: two point masses m connected by rigid, massless rods of length ℓ , with angles φ_1, φ_2 measured from the vertical ($-\hat{z}$) direction.

The transformation equations are

$$\vec{r}_1 = \ell(\sin \varphi_1 \hat{x} - \cos \varphi_1 \hat{z}) = \ell \hat{r}_1, \quad \vec{r}_2 = \ell(\sin \varphi_1 \hat{x} - \cos \varphi_1 \hat{z}) + \ell(\sin \varphi_2 \hat{x} - \cos \varphi_2 \hat{z}) = \ell(\hat{r}_1 + \hat{r}_{21}), \quad (33)$$

$$\hat{r}_1 = \sin \varphi_1 \hat{x} - \cos \varphi_1 \hat{z}, \quad \hat{r}_{21} = \sin \varphi_2 \hat{x} - \cos \varphi_2 \hat{z}. \quad (34)$$

The velocities follow from $\dot{\hat{r}}_1 = \dot{\varphi}_1 \partial \hat{r}_1 / \partial \varphi_1$ and $\dot{\hat{r}}_{21} = \dot{\varphi}_2 \partial \hat{r}_{21} / \partial \varphi_2$, with

$$\frac{\partial \hat{r}_1}{\partial \varphi_1} = \hat{e}_{\varphi_1} = \cos \varphi_1 \hat{x} + \sin \varphi_1 \hat{z}, \quad \frac{\partial \hat{r}_{21}}{\partial \varphi_2} = \hat{e}_{\varphi_2} = \cos \varphi_2 \hat{x} + \sin \varphi_2 \hat{z}, \quad (35)$$

so that

$$\dot{\vec{r}}_1 = \ell \dot{\hat{r}}_1 = \ell \dot{\varphi}_1 (\cos \varphi_1 \hat{x} + \sin \varphi_1 \hat{z}), \quad (36)$$

$$\dot{\vec{r}}_2 = \ell(\dot{\hat{r}}_1 + \dot{\hat{r}}_{21}) = \ell \dot{\varphi}_1 (\cos \varphi_1 \hat{x} + \sin \varphi_1 \hat{z}) + \ell \dot{\varphi}_2 (\cos \varphi_2 \hat{x} + \sin \varphi_2 \hat{z}). \quad (37)$$

The kinetic and potential energies are then

$$T = \frac{m}{2} [(\dot{\vec{r}}_1)^2 + (\dot{\vec{r}}_2)^2] = \frac{m}{2} [\ell^2 \dot{\varphi}_1^2 + \ell^2 (\dot{\varphi}_1^2 + \dot{\varphi}_2^2 + 2\dot{\varphi}_1 \dot{\varphi}_2 \hat{e}_{\varphi_1} \cdot \hat{e}_{\varphi_2})], \quad (38)$$

where

$$\hat{e}_{\varphi_1} \cdot \hat{e}_{\varphi_2} = \cos \varphi_1 \cos \varphi_2 + \sin \varphi_1 \sin \varphi_2 = \cos(\varphi_2 - \varphi_1), \quad (39)$$

$$T = \frac{m\ell^2}{2} [2\dot{\varphi}_1^2 + \dot{\varphi}_2^2 + 2\dot{\varphi}_1\dot{\varphi}_2 \cos(\varphi_2 - \varphi_1)], \quad V = mgz_1 + mgz_2 = -mg\ell (2 \cos \varphi_1 + \cos \varphi_2), \quad (40)$$

$$L = T - V = \frac{m\ell^2}{2} [2\dot{\varphi}_1^2 + \dot{\varphi}_2^2 + 2\dot{\varphi}_1\dot{\varphi}_2 \cos(\varphi_2 - \varphi_1)] + mg\ell (2 \cos \varphi_1 + \cos \varphi_2). \quad (41)$$

3.2 Small-Oscillation Expansion

The configuration $\varphi_1 = \varphi_2 = 0$ is known to be a stable equilibrium, but let us verify it directly:

$$\frac{\partial V}{\partial \varphi_1} = mg\ell \sin \varphi_1 = 0 \quad \text{at } \varphi_1 = 0 \checkmark, \quad \frac{\partial V}{\partial \varphi_2} = mg\ell \sin \varphi_2 = 0 \quad \text{at } \varphi_2 = 0 \checkmark, \quad (42)$$

and V is an absolute minimum at $\varphi_1 = \varphi_2 = 0$, so the equilibrium is stable. We therefore expand T and V to quadratic order in $\eta_i \equiv \varphi_i - \varphi_i^0 = \varphi_i$ (since $\varphi_i^0 = 0$):

$$\dot{\varphi}_1\dot{\varphi}_2 \cos(\varphi_2 - \varphi_1) = \dot{\eta}_1\dot{\eta}_2 \cos(\eta_2 - \eta_1) \approx \dot{\eta}_1\dot{\eta}_2 \left[1 - \frac{(\eta_2 - \eta_1)^2}{2} + \dots \right] \approx \dot{\eta}_1\dot{\eta}_2 + O(\dot{\eta}\dot{\eta} \eta^2) \quad (\eta_i \text{ small}), \quad (43)$$

$$\cos \varphi_i = \cos \eta_i = 1 - \frac{\eta_i^2}{2} + \frac{\eta_i^4}{24} - \dots \approx 1 - \frac{\eta_i^2}{2}. \quad (44)$$

Thus, to quadratic order in η and $\dot{\eta}$,

$$V \approx -mg\ell \left[3 - \frac{2\eta_1^2}{2} - \frac{\eta_2^2}{2} \right] = -mg\ell \left[3 - \eta_1^2 - \frac{1}{2}\eta_2^2 \right] \implies \boxed{\mathbf{V} = mg\ell \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}}, \quad (45)$$

$$T \approx \frac{m\ell^2}{2} [2\dot{\eta}_1^2 + \dot{\eta}_2^2 + 2\dot{\eta}_1\dot{\eta}_2] \implies \boxed{\mathbf{\Pi} = m\ell^2 \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}}. \quad (46)$$

As a check, $(\eta_1 \ \eta_2)\mathbf{V}(\eta_1 \ \eta_2)^T/(mg\ell) = 2\eta_1^2 + \eta_2^2$ and $(\dot{\eta}_1 \ \dot{\eta}_2)\mathbf{\Pi}(\dot{\eta}_1 \ \dot{\eta}_2)^T/(m\ell^2) = 2\dot{\eta}_1^2 + 2\dot{\eta}_1\dot{\eta}_2 + \dot{\eta}_2^2$, both consistent with Eqs. (45)–(46).

3.3 Normal Modes

To simplify the algebra, write $mg\ell = m\ell^2(g/\ell)$ and $\omega_0^2 \equiv g/\ell$. Then

$$0 = \text{Det} [\mathbf{V} - \omega^2 \mathbf{\Pi}] = \text{Det} \left(m\ell^2 \begin{pmatrix} 2\omega_0^2 - 2\omega^2 & -\omega^2 \\ -\omega^2 & \omega_0^2 - \omega^2 \end{pmatrix} \right) \quad (47)$$

$$\iff 0 = (2\omega_0^2 - 2\omega^2)(\omega_0^2 - \omega^2) - \omega^4 = 2\omega^4 - 4\omega^2\omega_0^2 + 2\omega_0^4 - \omega^4 = \omega^4 - 4\omega_0^2\omega^2 + 2\omega_0^4, \quad (48)$$

$$\iff 0 = (\omega^2 - 2\omega_0^2)^2 - \underbrace{(4\omega_0^4 - 2\omega_0^4)}_{2\omega_0^4} \quad (\text{completing the square}) \quad (49)$$

$$\iff \omega^2 = 2\omega_0^2 \pm \sqrt{2}\omega_0^2 = (2 \pm \sqrt{2})\omega_0^2 \iff \boxed{\omega = \begin{cases} \omega_+ = \sqrt{2 + \sqrt{2}}\omega_0 \\ \omega_- = \sqrt{2 - \sqrt{2}}\omega_0 \end{cases}} \quad (\text{eigenfrequencies}). \quad (50)$$

The eigenvector equation is

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2\omega_0^2 - 2\omega^2 & -\omega^2 \\ -\omega^2 & \omega_0^2 - \omega^2 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \quad (51)$$

$$0 = 2(\omega_0^2 - \omega^2)a_1 - \omega^2 a_2 \iff a_2 = 2 \left(\frac{\omega_0^2}{\omega^2} - 1 \right) a_1 \iff \frac{a_2}{a_1} = 2 \left(\frac{\omega_0^2}{\omega^2} - 1 \right), \quad (52)$$

$$0 = -\omega^2 a_1 + (\omega_0^2 - \omega^2)a_2 \iff a_1 = \left(\frac{\omega_0^2}{\omega^2} - 1 \right) a_2 \iff \frac{a_2}{a_1} = \frac{1}{\left(\frac{\omega_0^2}{\omega^2} - 1 \right)}. \quad (53)$$

Side note: the two equations for a_2/a_1 are equivalent,

$$2 \left(\frac{\omega_0^2}{\omega^2} - 1 \right) = \frac{1}{\left(\frac{\omega_0^2}{\omega^2} - 1 \right)} \iff 1 = 2 \left(\frac{\omega_0^2}{\omega^2} - 1 \right) \left(\frac{\omega_0^2}{\omega^2} - 1 \right) \quad (54)$$

$$\iff \omega^4 = 2(\omega_0^2 - \omega^2)(\omega_0^2 - \omega^2) \iff 0 = 2(\omega_0^2 - \omega^2)^2 - \omega^4 \iff 0 = \text{Det}(\mathbf{V} - \omega^2 \mathbf{\Pi}) \checkmark. \quad (55)$$

So either equation may be used, together with normalization, to get $\vec{a}$. Propose

$$\vec{a} = N \begin{pmatrix} 1 \\ 2 \left(\frac{\omega_0^2}{\omega^2} - 1 \right) \end{pmatrix}, \quad N : \text{normalization constant}, \quad x \equiv \frac{\omega_0^2}{\omega^2}. \quad (56)$$

Normalizing via $1 = \vec{a}^T \mathbf{\Pi} \vec{a}$,

$$\begin{aligned} 1 &= N^2 (1 \quad 2(x-1)) m \ell^2 \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2(x-1) \end{pmatrix} \\ &= N^2 m \ell^2 (1 \quad 2(x-1)) \begin{pmatrix} 2x \\ 2x-1 \end{pmatrix} = N^2 m \ell^2 [2x + 2(x-1)(2x-1)] \\ &= 2N^2 m \ell^2 [2x^2 - 3x + x + 1] = 2N^2 m \ell^2 [2x^2 - 2x + 1] \\ &= 2N^2 m \ell^2 [2(x-1)^2 + 4x - 2 + 1] = 2N^2 m \ell^2 \left[\underbrace{2(x-1)^2 - 1}_0 + 2x \right] = 4N^2 m \ell^2 \frac{\omega_0^2}{\omega^2}, \end{aligned} \quad (57)$$

using the identity $(x-1)^2 = x^2 - 2x + 1$, so that $2x^2 - 2x + 1 = 2(x-1)^2 + 2x - 1 = [2(x-1)^2 - 1] + 2x$, and $2(x-1)^2 - 1$ vanishes once $x = \omega_0^2/\omega^2$ is evaluated at either root of Eq. (50), since $\omega^4 [2(x-1)^2 - 1] = 2(\omega_0^2 - \omega^2)^2 - \omega^4 = 0$. Hence

$$N_{\pm} = \frac{1}{\sqrt{4m}} \frac{\omega_{\pm}}{\ell \omega_0} = \left(\frac{2 \pm \sqrt{2}}{4m \ell^2} \right)^{1/2}, \quad (58)$$

and, using $\frac{\omega_0^2}{\omega_{\pm}^2} = \frac{1}{2 \pm \sqrt{2}} = \frac{2 \mp \sqrt{2}}{4 - 2} = 1 \mp \frac{1}{\sqrt{2}}$, so that $2 \left(\frac{\omega_0^2}{\omega_{\pm}^2} - 1 \right) = \mp \sqrt{2}$,

$$\vec{a}_{\pm} = \left(\frac{2 \pm \sqrt{2}}{4m \ell^2} \right)^{1/2} \begin{pmatrix} 1 \\ \mp \sqrt{2} \end{pmatrix} \quad (59)$$

$$\omega_{\pm} = (2 \pm \sqrt{2})^{1/2} \sqrt{\frac{g}{\ell}}. \quad (60)$$

Writing the mode matrix as $\mathbf{A} = (\vec{a}_+ \vec{a}_-) = (4m\ell^2)^{-1/2} \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, comparison with $\vec{a}_\pm$ above gives

$$a \equiv (2 + \sqrt{2})^{1/2}, \quad c = -\sqrt{2}a = -\sqrt{2}(2 + \sqrt{2})^{1/2}, \quad (61)$$

$$b \equiv (2 - \sqrt{2})^{1/2}, \quad d = +\sqrt{2}b = +\sqrt{2}(2 - \sqrt{2})^{1/2}, \quad (62)$$

$$\boxed{\mathbf{A} = (\vec{a}_+ \vec{a}_-) = (4m\ell^2)^{-1/2} \begin{pmatrix} a & b \\ c & d \end{pmatrix}.} \quad (63)$$

Figure 5 shows the corresponding displacement patterns.

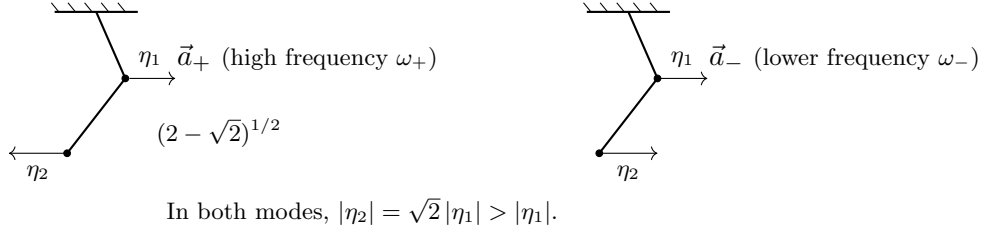


Figure 5: Displacement pattern of the two normal modes of the double pendulum: the high-frequency mode $\vec{a}_+$ and the low-frequency mode $\vec{a}_-$; in both, the lower mass has the larger angular displacement, $|\eta_2| = \sqrt{2}|\eta_1|$.

3.4 Normal Coordinates and Time Evolution

The normal coordinates are $\vec{\zeta} = \mathbf{A}^{-1}\vec{\eta} = \mathbf{A}^T\mathbf{\Pi}\vec{\eta}$, i.e.

$$\begin{aligned} \vec{\zeta} &= \mathbf{A}^T\mathbf{\Pi}\vec{\eta} = \mathbf{A}^T m\ell^2 \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = (4m\ell^2)^{-1/2} \begin{pmatrix} a & c \\ b & d \end{pmatrix} m\ell^2 \begin{pmatrix} 2\eta_1 + \eta_2 \\ \eta_1 + \eta_2 \end{pmatrix} \\ &= \left(\frac{m\ell^2}{4}\right)^{1/2} \begin{pmatrix} (2a + c)\eta_1 + (a + c)\eta_2 \\ (2b + d)\eta_1 + (b + d)\eta_2 \end{pmatrix}. \end{aligned} \quad (64)$$

Using $c = -\sqrt{2}a$ and $d = \sqrt{2}b$,

$$2a + c = (2 - \sqrt{2})a, \quad a + c = (1 - \sqrt{2})a, \quad 2b + d = (2 + \sqrt{2})b, \quad b + d = (1 + \sqrt{2})b, \quad (65)$$

and since $2 - \sqrt{2} = \sqrt{2}(\sqrt{2} - 1)$ and $1 - \sqrt{2} = -(\sqrt{2} - 1)$, the first row factors as $(\sqrt{2} - 1)a [\sqrt{2}\eta_1 - \eta_2]$; likewise $2 + \sqrt{2} = \sqrt{2}(\sqrt{2} + 1)$ and $1 + \sqrt{2} = (\sqrt{2} + 1)$, so the second row factors as $(\sqrt{2} + 1)b [\sqrt{2}\eta_1 + \eta_2]$. With $(m\ell^2/4)^{1/2} = m^{1/2}\ell/2$, $a = (2 + \sqrt{2})^{1/2}$, and $b = (2 - \sqrt{2})^{1/2}$, this gives

$$\boxed{\zeta_+(t) = m^{1/2}\ell \left(\frac{\sqrt{2} - 1}{2\sqrt{2}}\right)^{1/2} [\sqrt{2}\eta_1(t) - \eta_2(t)]} \quad (66)$$

$$\boxed{\zeta_-(t) = m^{1/2}\ell \left(\frac{\sqrt{2} + 1}{2\sqrt{2}}\right)^{1/2} [\sqrt{2}\eta_1(t) + \eta_2(t)]} \quad (67)$$

$$\dot{\zeta}_+(t) = m^{1/2}\ell \left(\frac{\sqrt{2}-1}{2\sqrt{2}} \right)^{1/2} \left[\sqrt{2} \dot{\eta}_1(t) - \dot{\eta}_2(t) \right], \quad (68)$$

$$\dot{\zeta}_-(t) = m^{1/2}\ell \left(\frac{\sqrt{2}+1}{2\sqrt{2}} \right)^{1/2} \left[\sqrt{2} \dot{\eta}_1(t) + \dot{\eta}_2(t) \right]. \quad (69)$$

But we also know that, being normal coordinates,

$$\zeta_+(t) = f_+ \cos(\omega_+ t + \delta_+), \quad \zeta_-(t) = f_- \cos(\omega_- t + \delta_-), \quad (70)$$

$$\dot{\zeta}_+(t) = -f_+ \omega_+ \sin(\omega_+ t + \delta_+), \quad \dot{\zeta}_-(t) = -f_- \omega_- \sin(\omega_- t + \delta_-). \quad (71)$$

The Lagrangian in terms of the normal coordinates is, from Eq. (50),

$$\begin{aligned} L &= \frac{1}{2} \sum_{\alpha} \left(\dot{\zeta}_{\alpha}^2 - \omega_{\alpha}^2 \zeta_{\alpha}^2 \right) = \frac{1}{2} \left[\left(\dot{\zeta}_+^2 - \omega_+^2 \zeta_+^2 \right) + \left(\dot{\zeta}_-^2 - \omega_-^2 \zeta_-^2 \right) \right] \\ &= \frac{1}{2} \left[\left(\dot{\zeta}_+^2 - \left[(2 + \sqrt{2}) \frac{g}{\ell} \right] \zeta_+^2 \right) + \left(\dot{\zeta}_-^2 - \left[(2 - \sqrt{2}) \frac{g}{\ell} \right] \zeta_-^2 \right) \right]. \end{aligned} \quad (72)$$

3.5 Initial Conditions and Explicit Time Dependence

Assume the initial condition that at $t = 0$: $\varphi_1 = \varphi_2 = 0$ and $\dot{\varphi}_1 = -\dot{\varphi}_2 = v > 0$; we seek $\varphi_1(t), \varphi_2(t)$.

From $\zeta_+(0) = 0 = f_+ \cos \delta_+$ and $\zeta_-(0) = 0 = f_- \cos \delta_-$, we need $\delta_+ = \pm \frac{\pi}{2}, \pm \frac{3}{2}\pi, \dots$ and $\delta_- = \pm \frac{\pi}{2}, \pm \frac{3}{2}\pi, \dots$. Using Eq. (67) at $t = 0$ (with $\eta_1 = \eta_2 = 0, \dot{\eta}_1 = v, \dot{\eta}_2 = -v$),

$$-f_+ \omega_+ \sin \delta_+ = \dot{\zeta}_+(0) = m^{1/2}\ell \left(\frac{\sqrt{2}-1}{2\sqrt{2}} \right)^{1/2} \left[\sqrt{2} v - (-v) \right] = \sqrt{2 + \sqrt{2}} \frac{m^{1/2}\ell v}{2}, \quad (73)$$

$$-f_- \omega_- \sin \delta_- = \dot{\zeta}_-(0) = m^{1/2}\ell \left(\frac{\sqrt{2}+1}{2\sqrt{2}} \right)^{1/2} \left[\sqrt{2} v + (-v) \right] = \sqrt{2 - \sqrt{2}} \frac{m^{1/2}\ell v}{2}. \quad (74)$$

Choosing $\delta_+ = \delta_- = -\pi/2$ (so $-\sin \delta_{\pm} = 1$) and using $\omega_{\pm} = \sqrt{2 \pm \sqrt{2}} \sqrt{g/\ell}$,

$$f_{\pm} \sqrt{2 \pm \sqrt{2}} \sqrt{\frac{g}{\ell}} = \sqrt{2 \pm \sqrt{2}} \frac{m^{1/2}\ell v}{2} \implies \boxed{f_{\pm} = \frac{v}{2} \left[\frac{m\ell^3}{g} \right]^{1/2}}. \quad (75)$$

With $\delta_{\pm} = -\pi/2$, $\cos(\omega t - \pi/2) = \cos \omega t \cos \frac{\pi}{2} + \sin \omega t \sin \frac{\pi}{2} = \sin \omega t$, so

$$\zeta_{\pm}(t) = f_{\pm} \cos\left(\omega_{\pm} t - \frac{\pi}{2}\right) = f_{\pm} \sin(\omega_{\pm} t) \implies \boxed{\zeta_{\pm}(t) = \frac{v}{2} \left[\frac{m\ell^3}{g} \right]^{1/2} \sin\left(\left[(2 \pm \sqrt{2}) \frac{g}{\ell}\right]^{1/2} t\right)}. \quad (76)$$

In terms of the original coordinates, $\vec{\eta}(t) = \mathbf{A}\vec{\zeta}(t)$, i.e., using Eq. (63),

$$\vec{\eta}(t) = \mathbf{A}\vec{\zeta}(t) = (4m\ell^2)^{-1/2} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \zeta_+(t) \\ \zeta_-(t) \end{pmatrix} = (4m\ell^2)^{-1/2} \begin{pmatrix} a\zeta_+(t) + b\zeta_-(t) \\ c\zeta_+(t) + d\zeta_-(t) \end{pmatrix}. \quad (77)$$

Substituting $a = (2 + \sqrt{2})^{1/2}$, $b = (2 - \sqrt{2})^{1/2}$, $c = -\sqrt{2}a$, $d = \sqrt{2}b$, and $(4m\ell^2)^{-1/2} \frac{v}{2} [m\ell^3/g]^{1/2} = \frac{v}{4\sqrt{g/\ell}}$, together with Eq. (76), gives

$$\begin{aligned} \varphi_1(t) = \eta_1(t) &= \frac{v}{4\sqrt{g/\ell}} \left\{ (2 + \sqrt{2})^{1/2} \sin \left(\left[(2 + \sqrt{2}) \frac{g}{\ell} \right]^{1/2} t \right) \right. \\ &\quad \left. + (2 - \sqrt{2})^{1/2} \sin \left(\left[(2 - \sqrt{2}) \frac{g}{\ell} \right]^{1/2} t \right) \right\}, \\ \varphi_2(t) = \eta_2(t) &= \frac{\sqrt{2}v}{4\sqrt{g/\ell}} \left\{ -(2 + \sqrt{2})^{1/2} \sin \left(\left[(2 + \sqrt{2}) \frac{g}{\ell} \right]^{1/2} t \right) \right. \\ &\quad \left. + (2 - \sqrt{2})^{1/2} \sin \left(\left[(2 - \sqrt{2}) \frac{g}{\ell} \right]^{1/2} t \right) \right\}. \end{aligned} \quad (78)$$

Differentiating,

$$\begin{aligned} \dot{\varphi}_1(t) &= \frac{v}{4} \left\{ (2 + \sqrt{2}) \cos \left(\left[(2 + \sqrt{2}) \frac{g}{\ell} \right]^{1/2} t \right) + (2 - \sqrt{2}) \cos \left(\left[(2 - \sqrt{2}) \frac{g}{\ell} \right]^{1/2} t \right) \right\}, \\ \dot{\varphi}_2(t) &= \frac{\sqrt{2}v}{4} \left\{ -(2 + \sqrt{2}) \cos \left(\left[(2 + \sqrt{2}) \frac{g}{\ell} \right]^{1/2} t \right) + (2 - \sqrt{2}) \cos \left(\left[(2 - \sqrt{2}) \frac{g}{\ell} \right]^{1/2} t \right) \right\}. \end{aligned} \quad (79)$$

As a check, at $t = 0$ all $\sin(\dots)$ terms in Eq. (78) vanish, giving $\varphi_1 = \varphi_2 = 0$ ✓; and at $t = 0$ all $\cos(\dots)$ terms in Eq. (79) equal 1, giving

$$\dot{\varphi}_1 = \frac{v}{4} \left\{ (2 + \sqrt{2}) + (2 - \sqrt{2}) \right\} = v \text{ ✓}, \quad \dot{\varphi}_2 = \frac{\sqrt{2}v}{4} \left\{ -(2 + \sqrt{2}) + (2 - \sqrt{2}) \right\} = -v \text{ ✓}. \quad (80)$$

4 Comparison with Quantum Mechanics

In quantum mechanics, an initial state is expanded in the energy eigenbasis $\{|\varphi_\alpha\rangle\}$,

$$|\psi(0)\rangle = \sum_{\alpha} c_{\alpha} |\varphi_{\alpha}\rangle, \quad c_{\alpha} = \langle \varphi_{\alpha} | \psi(0) \rangle, \quad (81)$$

and each coefficient evolves with its own phase,

$$|\psi(0)\rangle \rightarrow \{c_{\alpha}\} \rightarrow c_{\alpha} e^{-iE_{\alpha}t/\hbar} \rightarrow |\psi(t)\rangle = \sum_{\alpha} c_{\alpha} e^{-iE_{\alpha}t/\hbar} |\varphi_{\alpha}\rangle. \quad (82)$$

The small-oscillations problem proceeds in exact analogy:

$$\vec{\eta}(0), \dot{\vec{\eta}}(0) \longrightarrow \vec{\zeta}(0), \dot{\vec{\zeta}}(0) \longrightarrow \vec{\zeta}(t), \dot{\vec{\zeta}}(t) \longrightarrow \vec{\eta}(t), \dot{\vec{\eta}}(t), \quad (83)$$

$$\vec{\eta}(t) = \mathbf{A} \vec{\zeta}(t), \quad \vec{\zeta}(t) = \mathbf{A}^T \mathbf{\Pi} \dot{\vec{\eta}}(t). \quad (84)$$